

### Question 1

Not yet answered

Marked out of 2.00

Flag question

One of the following statements is incorrect

Select one:

- ☐ a. PLCs can withstand harsh industrial environment conditions
- ☐ b. All PLCs have an LCD screen connected to them
- ☐ c. PLCs are general purpose devices
- ☐ d. All PLCs don't have CD Drives

## Question 2

Not yet answered

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Flag question

LAN can be used with industrial networks covering a whole large city

Select one:

- ☐ a. False
- ☐ b. True



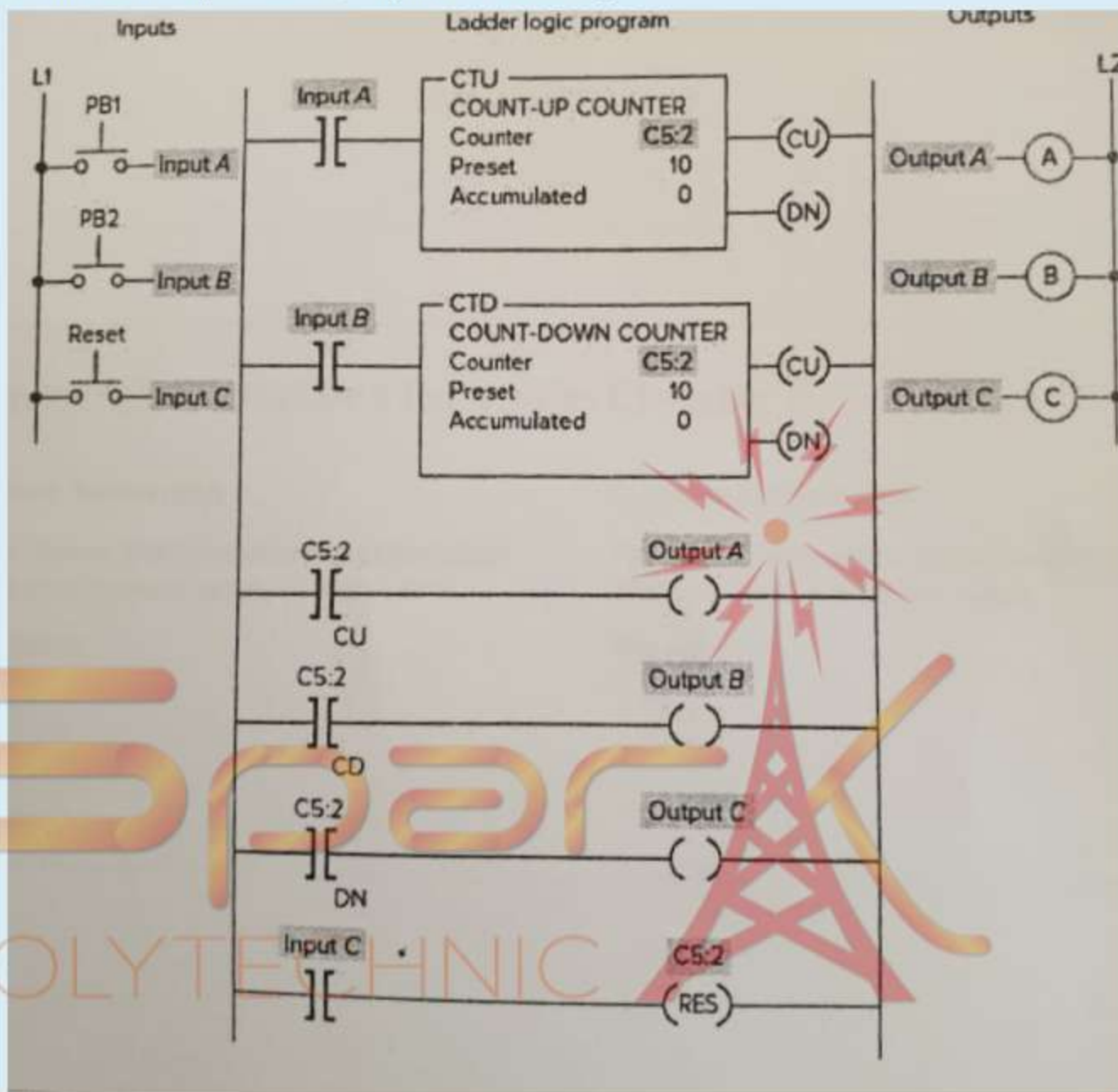
### Question 3

Not yet answered

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Flag question

For the ladder diagram below, **Output A** becomes high if:



Select one:

- ☐ a. None
- ☐ b. input A is activated 10 times regardless of input B
- ☐ c. input A is activated
- ☐ d. input A is activated 10 times more than input B, and Input C is not activated

**Question 4**

Not yet answered

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Flag question

In an industrial plant, it is required to automatically control a tank level by keeping it within the range (20 to 80)%. For this control loop, we may use:

Select one:

- ☐ a. Open loop control
- ☐ b. ON-OFF control
- ☐ c. PID Control
- ☐ d. None



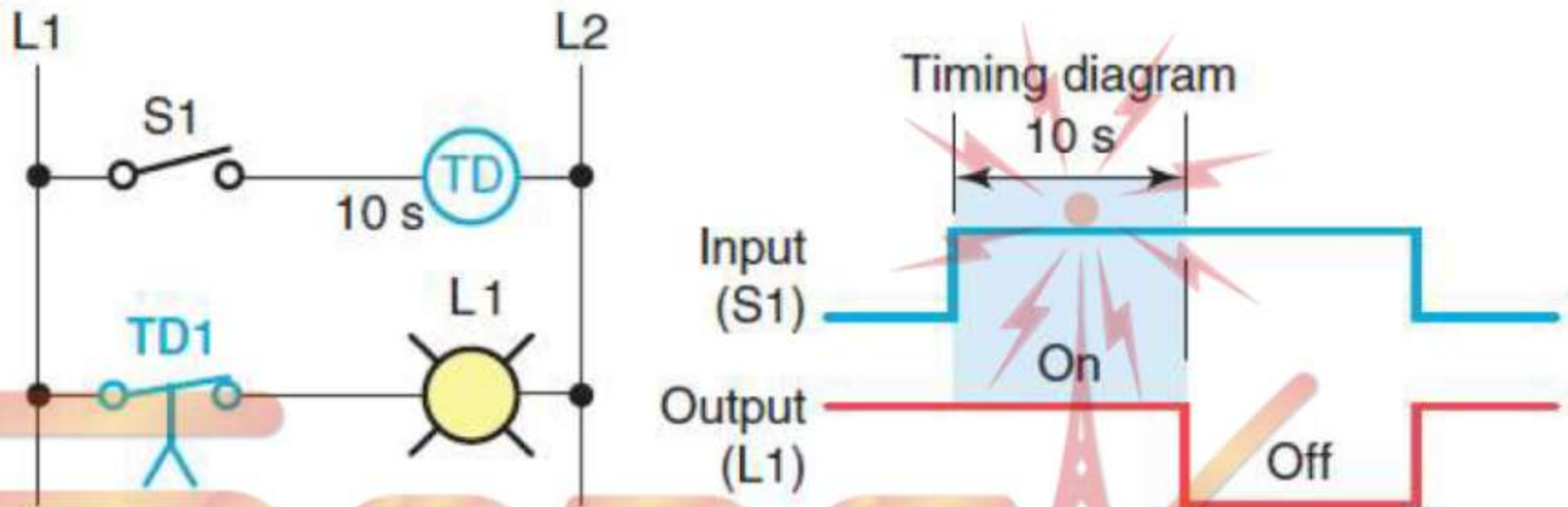
**Question 5**

Not yet answered

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The timing diagram shown below represents the behavior of



Select one:

- ☐ a. Up Down Counter
- ☐ b. Up Counter
- ☐ c. Off delay timer
- ☐ d. On delay timer



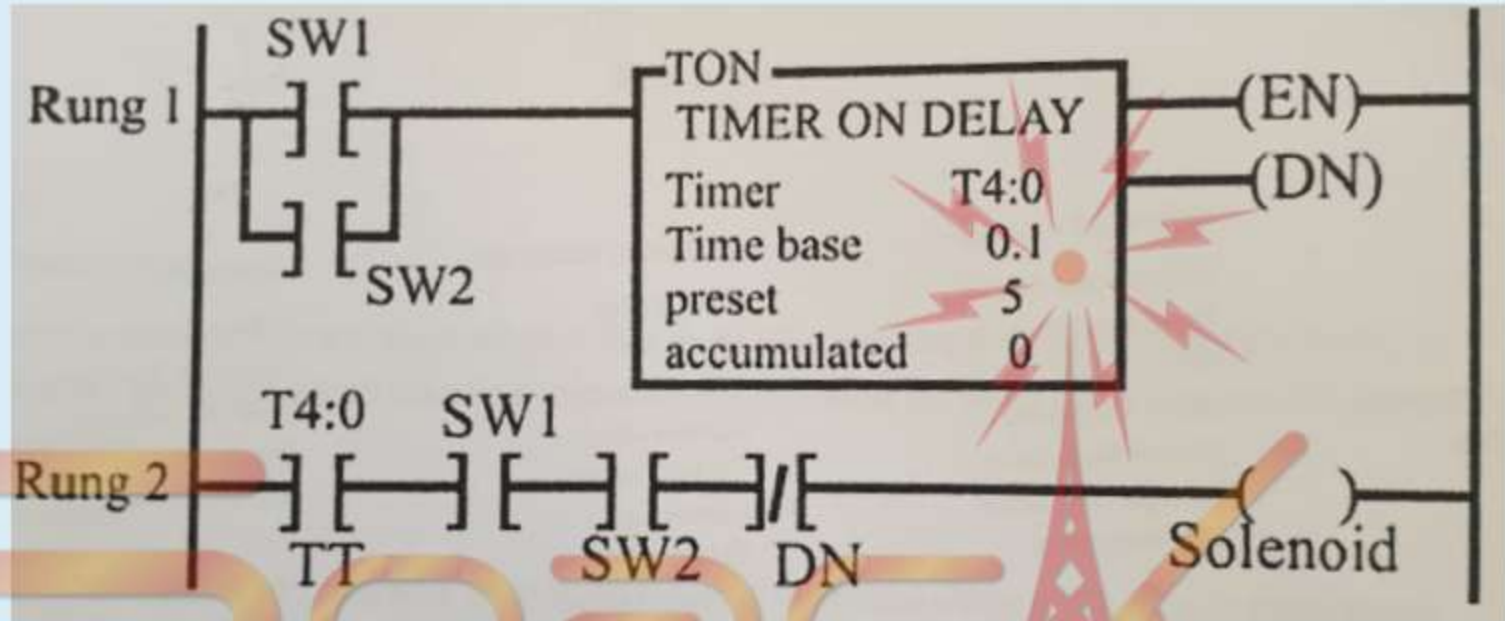
### Question 6

Not yet answered

Marked out of 2.00

Flag question

For the ladder below, the solenoid will be activated:



Select one:

- ☐ a. for 0.5 second after activating both SW1 and SW2
- ☐ b. for 5 seconds after activating both SW1 and SW2
- ☐ c. for 0.1 second after activating both SW1 and SW2
- ☐ d. for 0.5 second after activating SW1 or SW2

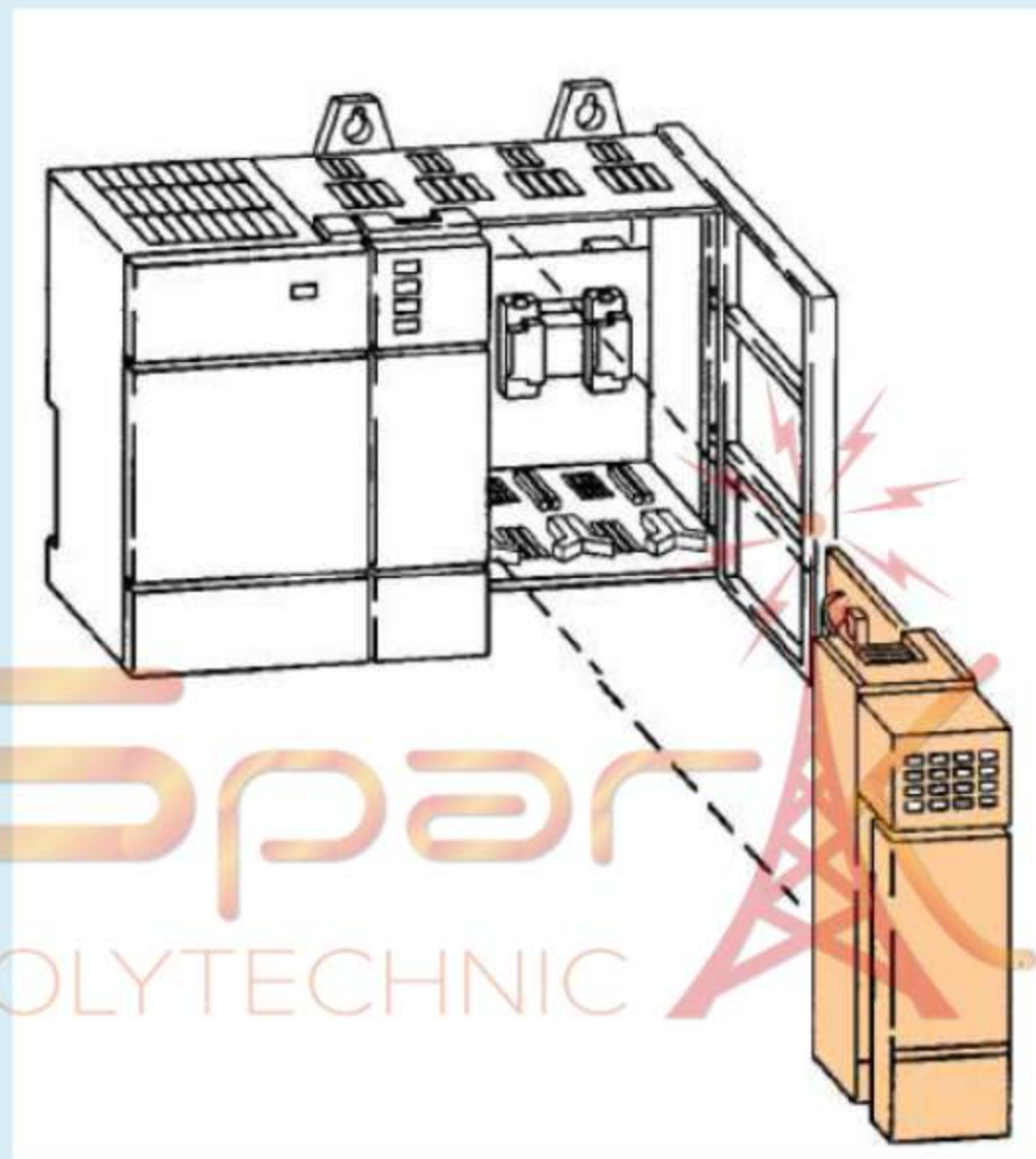
**Question 7**

Not yet answered

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**The figure below shows a**



Select one:

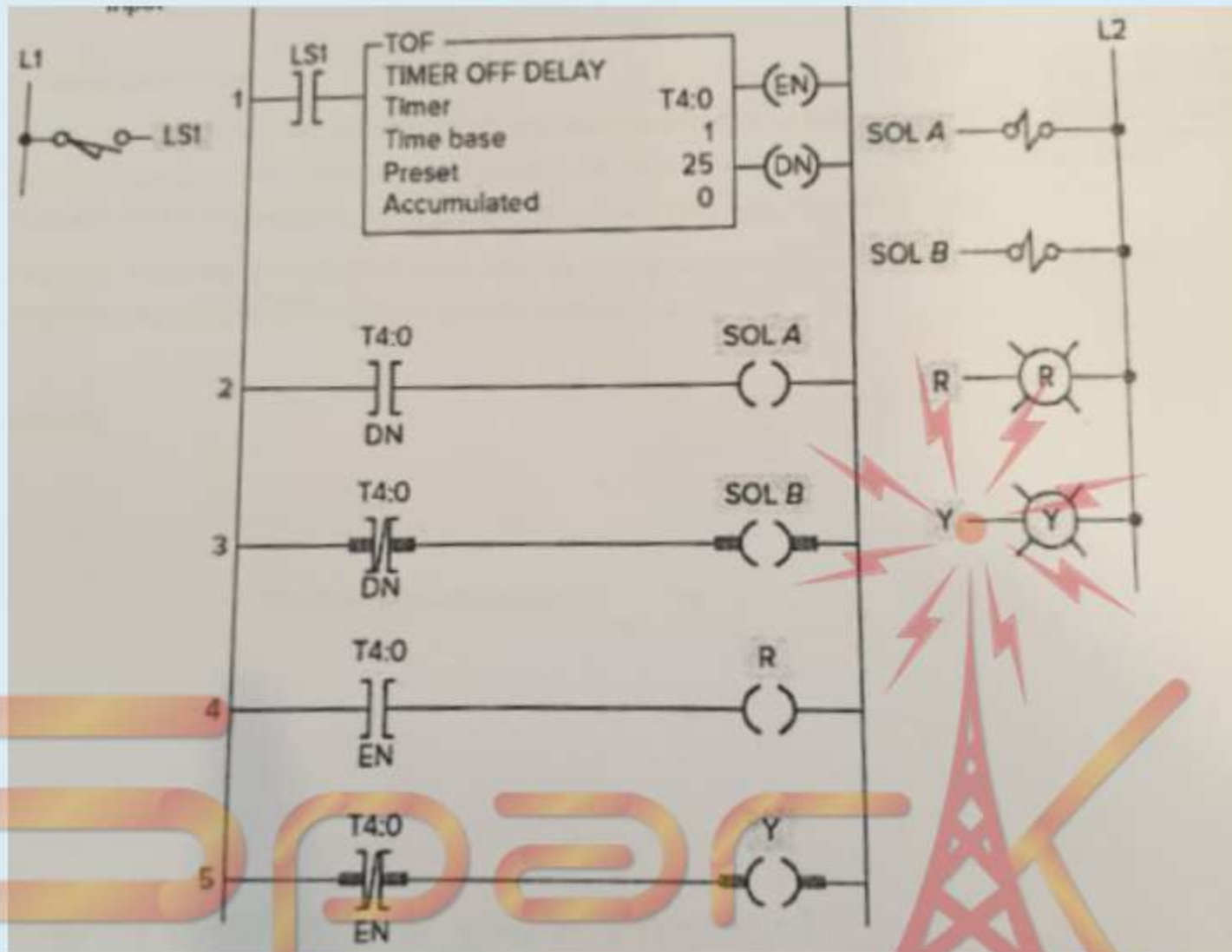
- ☐ a. Single board computer
- ☐ b. Modular input output PLC
- ☐ c. Fixed input output PLC
- ☐ d. Relay-based controller

**Question 8**

Not yet answered

Marked out of 2.00

Flag question

For the ladder diagram below, the output **R** will be high:

Select one:

- ☐ a. before activating LS1
- ☐ b. as long as LS1 is low
- ☐ c. as long as LS1 is high



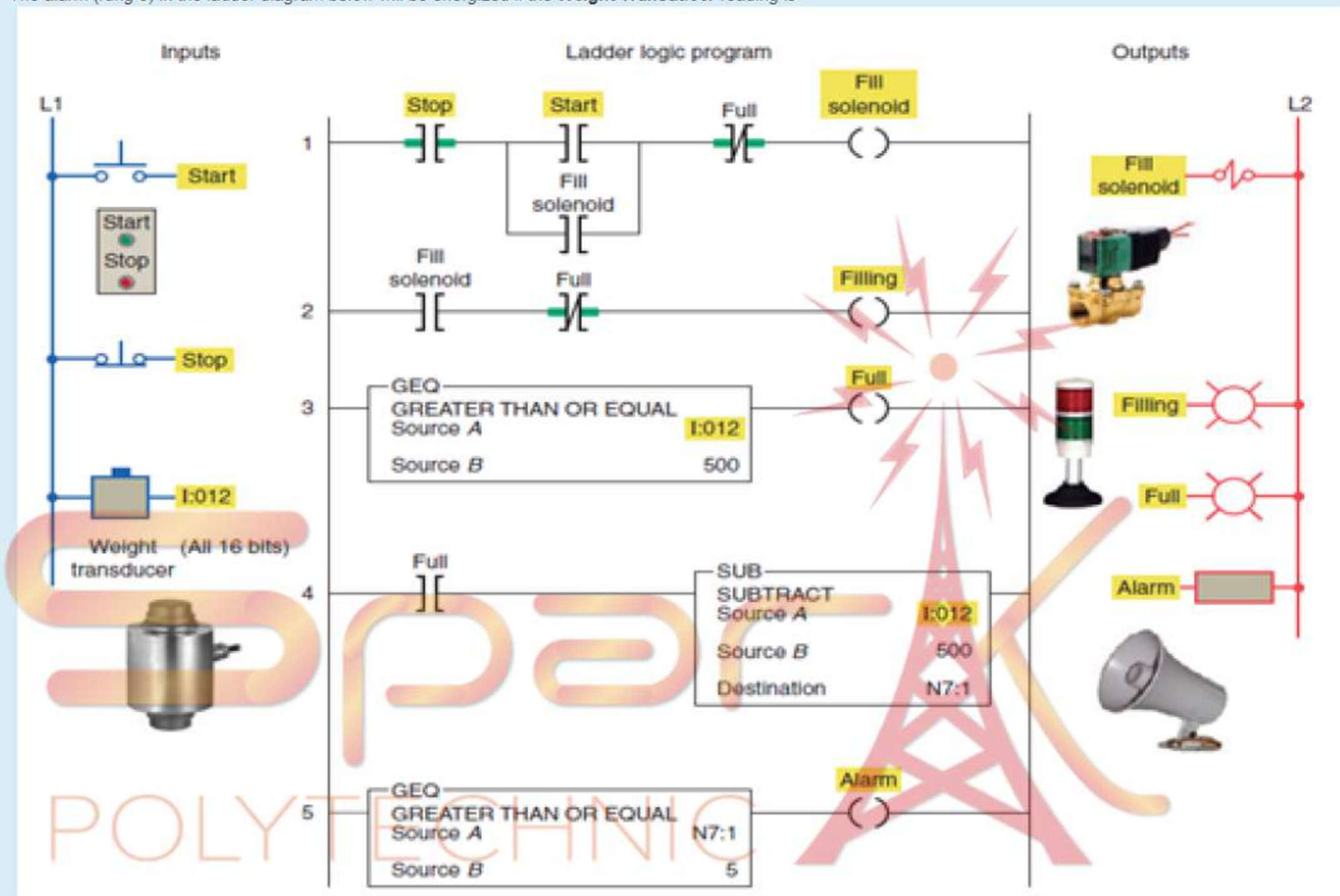
Question 9

Not yet answered

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The alarm (rung 5) in the ladder diagram below will be energized if the **Weight Transducer** reading is



Select one:

- ☐ a. Greater than 495 and less than 500
- ☐ b. Greater than or equal to 505
- ☐ c. Over 500
- ☐ d. Less than 500

**Question 10**

Not yet answered

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In a PLC ladder diagram, an Off delay timer (time base is 0.1 sec and preset value =50) is enabled through normally open switch S2. The TT bit will be enabled:

Select one:

- ☐ a. After 5 seconds of continuous activation of S2
- ☐ b. Directly when S2 is enabled
- ☐ c. Whenever the accumulated value of the timer is changing
- ☐ d. After 50 seconds of continuous activation of S2



### Question 11

Not yet answered

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For the statement list below, the output Q2.0 will be activated:

LD I0.0

LD I0.1

LD I0.2

CTUD C2, 10

LD C2

= Q2.0

Select one:

- ☐ a. If I0.2 is not activated and I0.0 is activated 10 times regardless of the state of I0.1.
- ☐ b. If I0.2 is activated 8 times more than I0.1
- ☐ c. If I0.2 is not activated and I0.0 is activated 10 times more than the I0.1.
- ☐ d. If I0.0 is activated 10 times more than I0.1 regardless of I0.2.



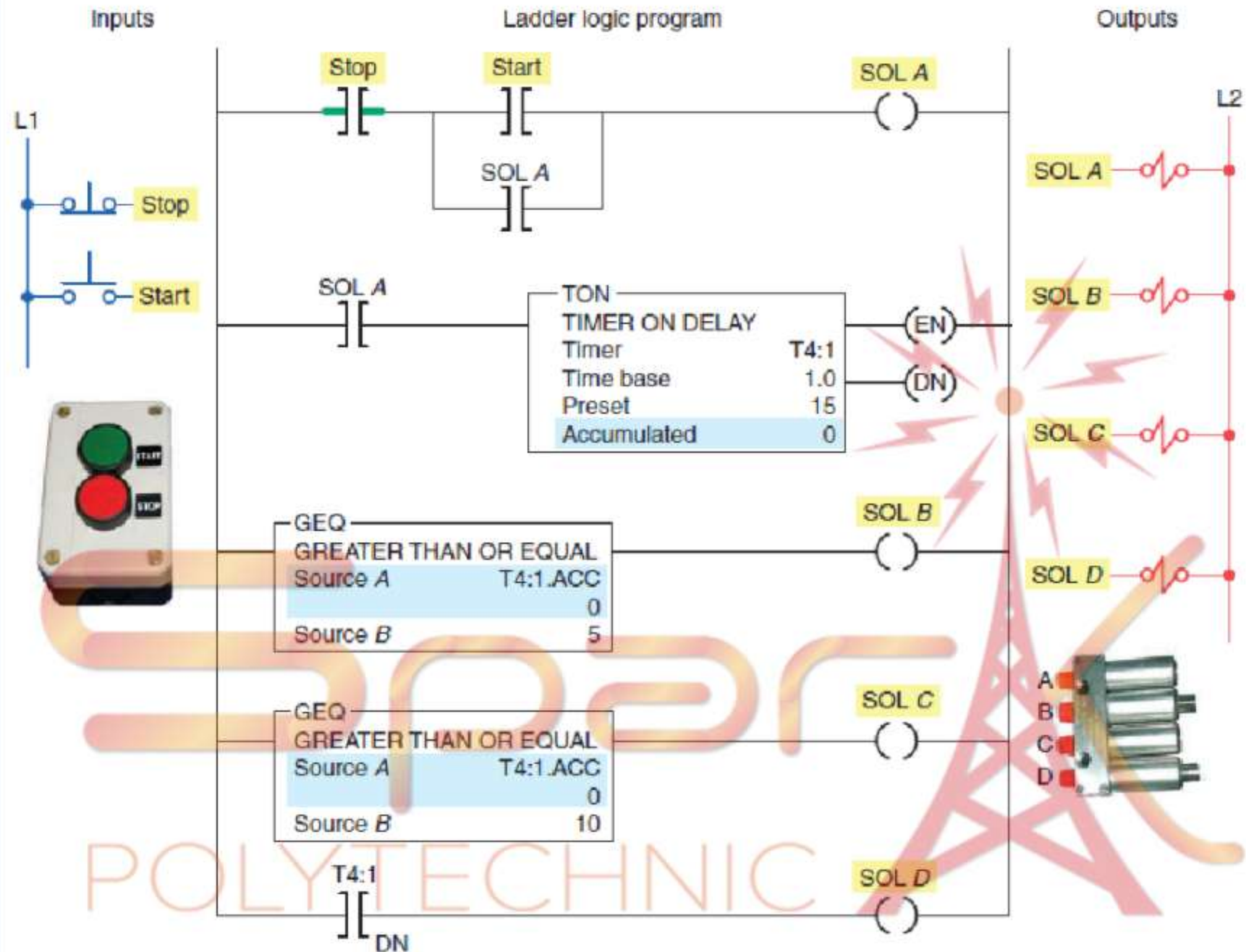
Question 12

Not yet answered

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For the following ladder diagram, when start pushbutton is pressed SOL A will be energized:



Select one:

- ☐ a. after 20 seconds
- ☐ b. after 10 seconds
- ☐ c. after 15 seconds
- ☐ d. immediately

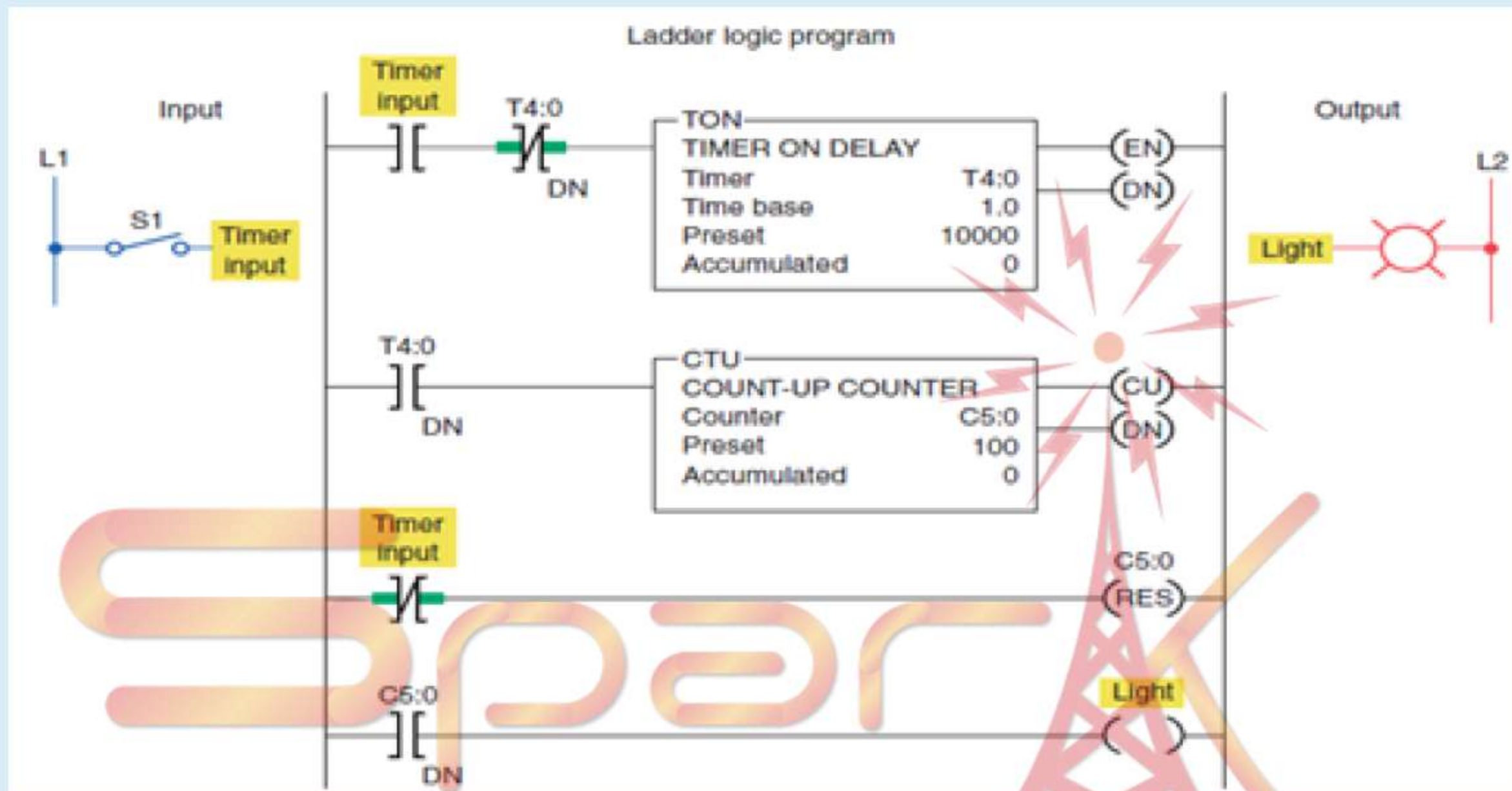
Question 13

Not yet answered

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For the following ladder, the accumulated value of the counter will be increased by one if



Select one:

- ☐ a. The timer input is activated for 100 seconds
- ☐ b. The timer input is activated for 10100 seconds
- ☐ c. The timer input is activated for 10000 seconds
- ☐ d. The timer input is activated for 1000 000 seconds



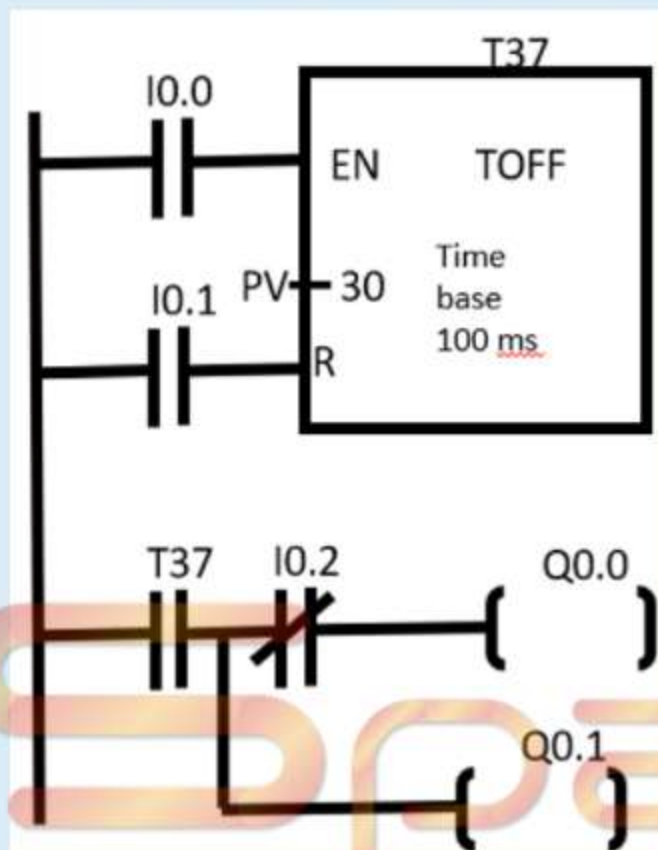
**Question 14**

Not yet answered

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For the following ladder diagram I0.1 and I0.2 are not activated at all, the output Q0.0 will be activated:



Select one:

- ☐ a. None
- ☐ b. After 0.3 seconds of activating I0.0
- ☐ c. After 3 seconds of activating I0.0
- ☐ d. Directly when I0.0 is activated

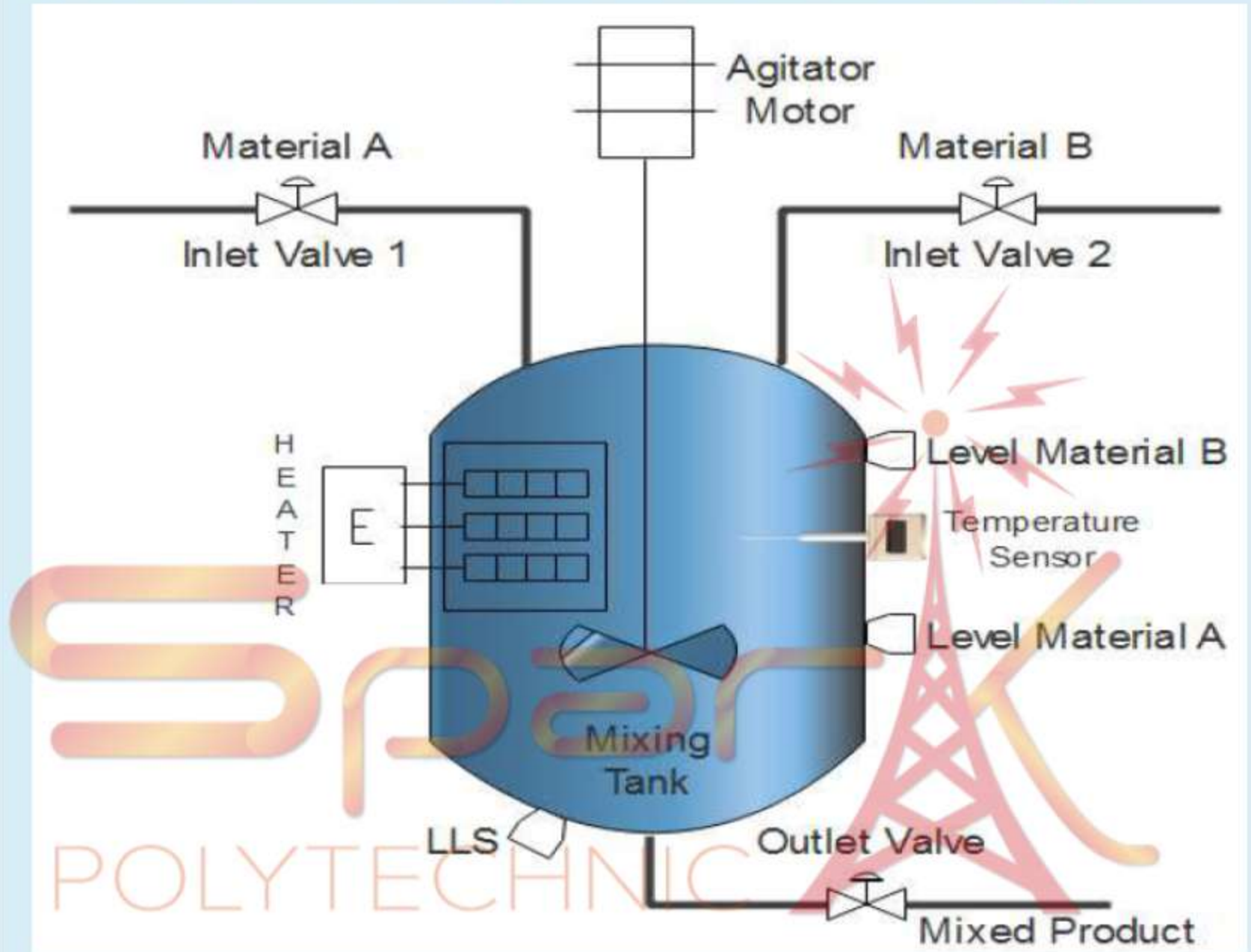
**Question 15**

Not yet answered

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For the process shown below, the following field devices should be connected to the output module of the PLC.



Select one:

- ☐ a. Inlet valve 1, Inlet valve 2, Low level Switch (LLS), motor agitator, outlet valve and temperature sensor
- ☐ b. Inlet valve 1, Inlet valve 2, Low level Switch (LLS), motor agitator and temperature sensor
- ☐ c. Inlet valve 1, Inlet valve 2, Heater, motor agitator and temperature sensor
- ☐ d. Inlet valve 1, inlet valve 2, Heater, motor agitator and outlet valve

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**Question 16**

Not yet answered

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One of the following is a factor that affecting the memory size needed for a particular PLC installation:

Select one:

- ☐ a. Programming device used
- ☐ b. PLC price
- ☐ c. Size of control program
- ☐ d. PLC manufacturer



### Question 17

Not yet answered

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The statement list below can be expressed as a logic statement as follows:

LD A

A B

LD C

A D

OLD

= Q0.0

Select one:

- ☐ a.  $Q0.0 = AB + CD$
- ☐ b.  $Q0.0 = A(B + C) + D$
- ☐ c.  $Q0.0 = (A + B)(C + D)$
- ☐ d.  $ABCD$



**Question 18**

Not yet answered

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The statement list below can be expressed as a logic statement as follows:

LDA

☐ B

LD C

☐ D

ALD

= Q0.0

Select one:

☐ a.  $Q0.0 = (A+B)(C+D)$

☐ b.  $Q0.0 = AB+CD$

☐ c.  $Q0.0 = A(B+C) + D$

☐ d. None

### Question 19

Not yet answered

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For the statement list below the output Q0.2 is high if

LDN I0.0

AN I0.1

LDN I0.1

AN I0.0

OLD

= Q0.2

Select one:

- ☐ a. I0.0 is high and whatever I0.1 is.
- ☐ b. I0.0 is high and I0.1 is high
- ☐ c. I0.0 is high and I0.1 is low
- ☐ d. I0.0 is low and I0.1 is low

**Question 20**

Not yet answered

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Input/output data files usually account for most of the memory of a PLC system.

Select one:

- ☐ a. True
- ☐ b. False



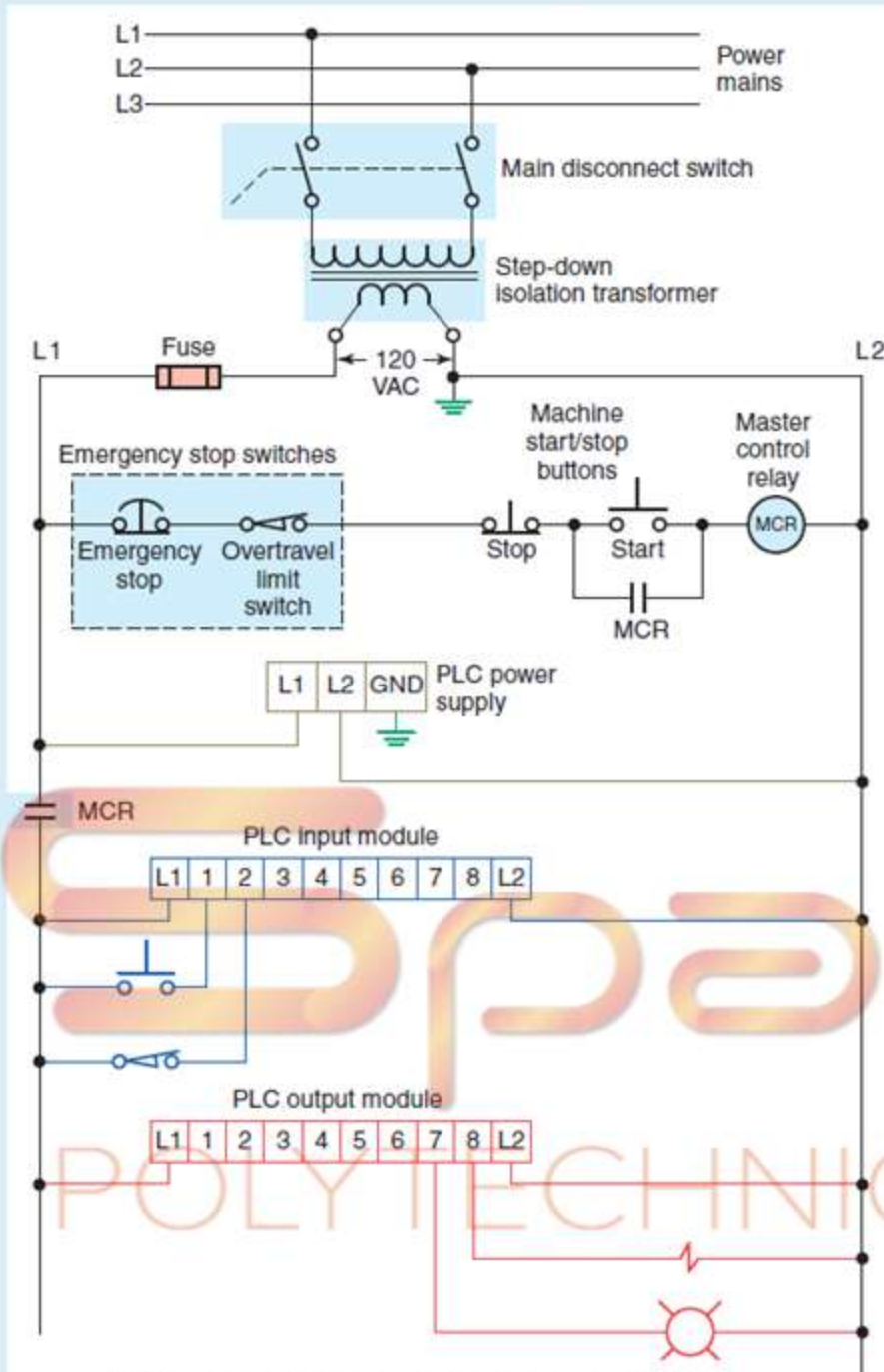
## Question 21

Not yet answered

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Consider the PLC wiring diagram below. The power main disconnect switch should be mounted



Safety wiring requirements for a PLC installation.  
Source: Courtesy Minarik Automation & Control.



Select one:

- ☐ a. in the control room
- ☐ b. None
- ☐ c. in the motor control center
- ☐ d. outside of the PLC enclosure



**Question 22**

Not yet answered

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If the content of the memory address AC0 is 16 and the content of the memory address AC1 is 64 before execution the PLC program below. Then:

**LD I0.0**

**SQRT AC0, AC1**

Select one:

- ☐ a. If I0.0 is activated, then the contents of AC0 is 4 and AC1 is 16
- ☐ b. If I0.0 is activated, then the contents of AC0 is 4 and AC1 is 4
- ☐ c. If I0.0 is activated, then the contents of AC0 is 16 and AC1 is 16
- ☐ d. If I0.0 is activated, then the contents of AC0 is 16 and AC1 is 4

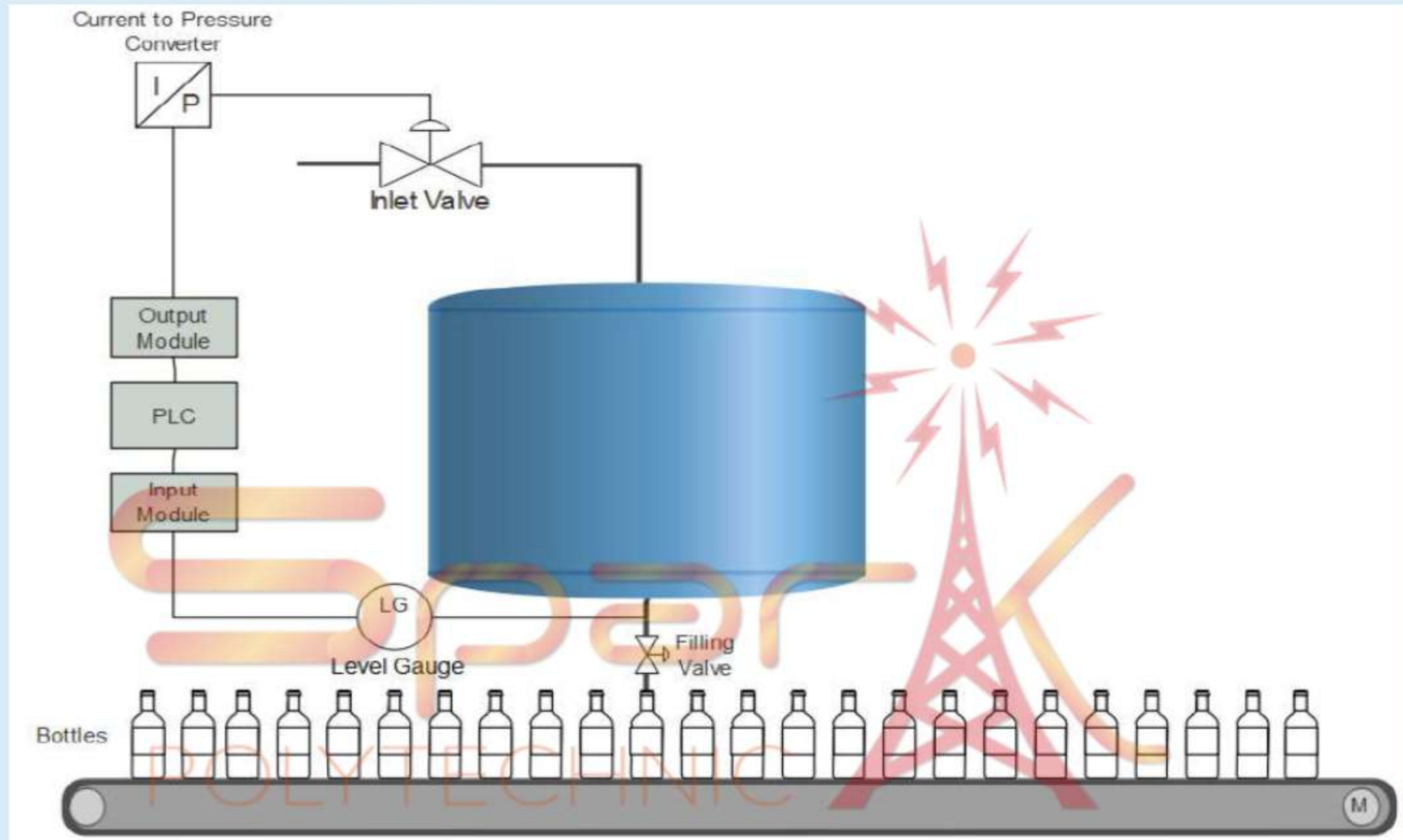
**Question 23**

Not yet answered

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The inlet valve in the PLC system below is expected to



Select one:

- ☐ a. receives a 4-20 mA signal from the I/P device
- ☐ b. receive an On-Off signal from the I/P and adjusts its opening accordingly
- ☐ c. behave as an On-Off actuator
- ☐ d. receive a variable signal from the I/P and adjusts its opening accordingly

### Question 24

Not yet answered

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The main disconnect switch is used in PLC circuits to connect or disconnect.....

Select one:

- ☐ a. Only the PLC power supply
- ☐ b. Only input devices
- ☐ c. Only the power of input/output modules.
- ☐ d. Both the input/output modules and the PLC power supply.

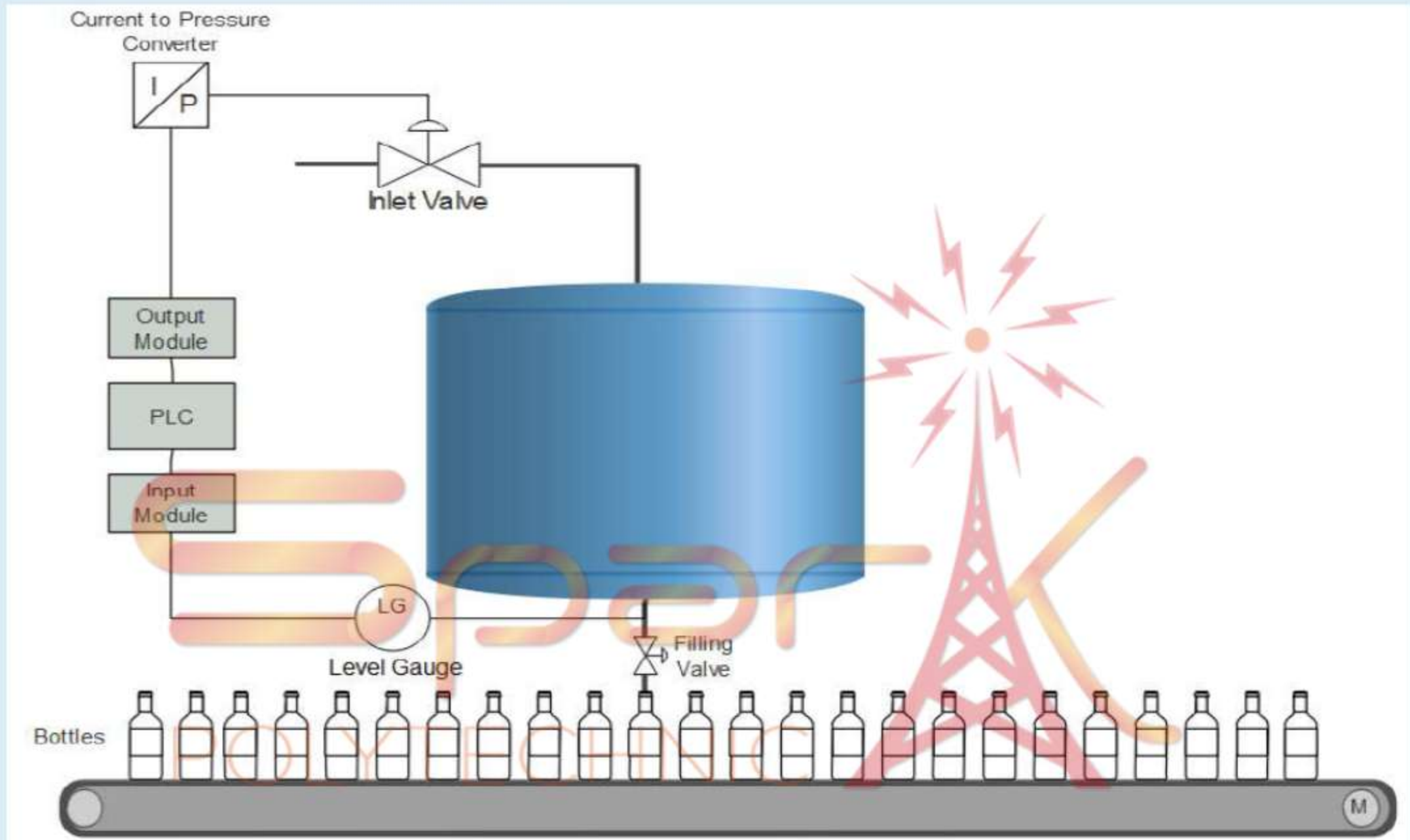
**Question 25**

Not yet answered

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Flag question

In the system shown below, the current to pressure converter is



Select one:

- ☐ a. an analog input analog output device
- ☐ b. a digital input digital output device
- ☐ c. a digital input analog output device
- ☐ d. an analog input digital output device